



Safe Biomedical Natural Language Inference for Clinical Trials

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Abstract

Our study evaluates the performance of various LLMs on the NLI task on Clinical trial data. we experimented with two strategies: Chain-of-Thought prompting, which encourages LLMs to generate step-by-step reasoning, and Zero-Shot prompting, which relies on the models' general language understanding capabilities without task-specific fine-tuning. We have also experimented different approaches for evidence retrieval.

Datasets

NLI4CT dataset is used

Clinical Trial Report

Intervention:.....
Eligibility:
Inclusion Criteria:....
Exclusion Criteria:

- Prior treatment with a phosphatidylinositol 3-kinase (PI3K) inhibitor, mammalian target of rapamycin (mTOR) inhibitor (e.g. everolimus), or protein kinase B (AKT) inhibitor

Results:.....
Adverse Events:

Adverse Events 1

- Total: 19/213 (8.92%)
- Diarrhea: 0/213 (0%)

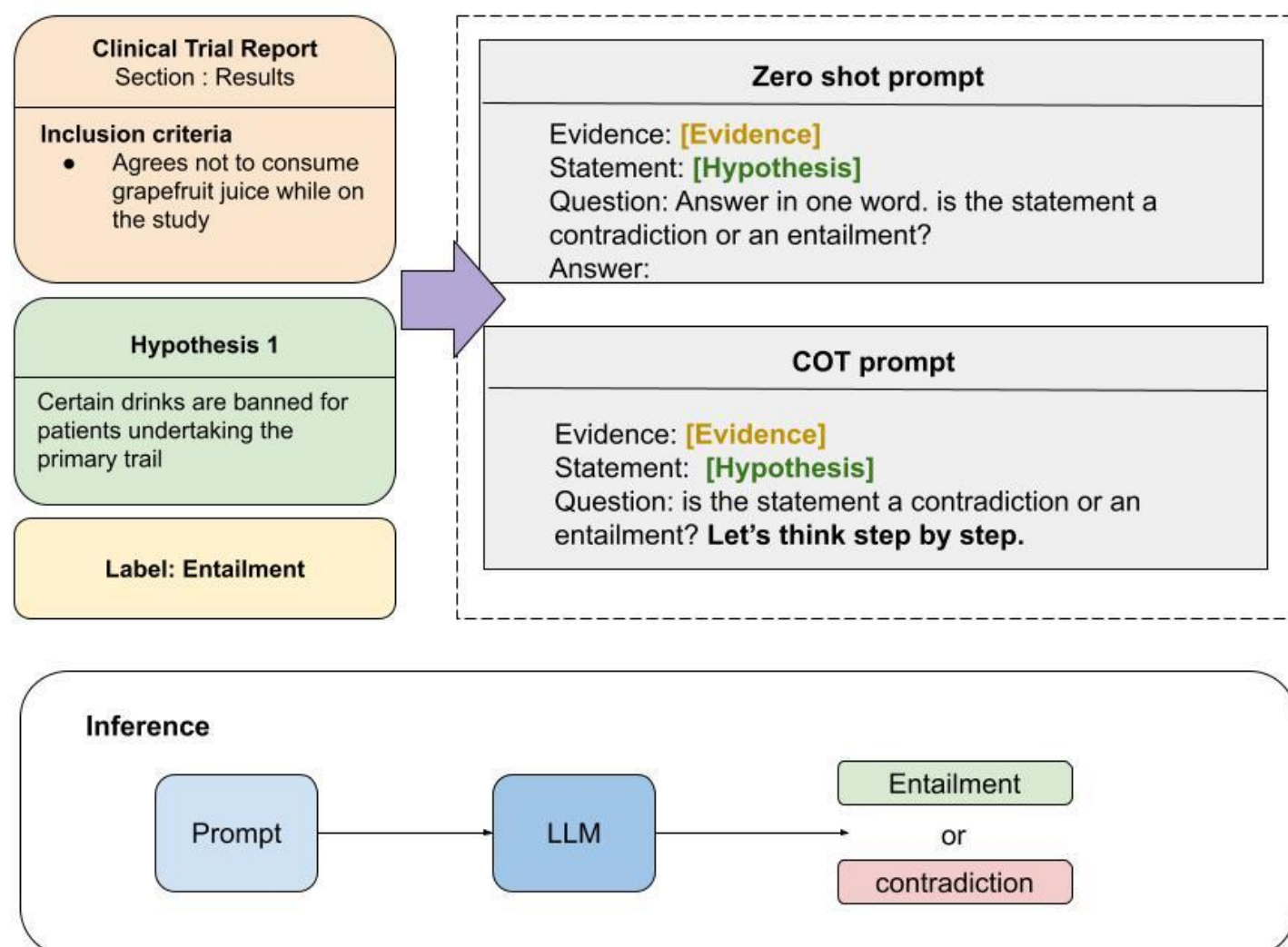
Adverse Events 2

- Total: 133/416 (31.97%)
- Diarrhea: 32/416 (7.69%)

Hypotheses:

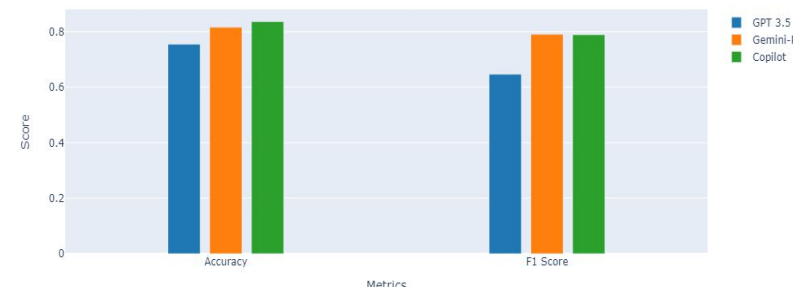
- A total of 32 patients in the primary trial had **diarrhoea**
- Prior treatment with fulvestrant or with a phosphatidylinositol 3-kinase (PI3K) inhibitor will not prevent a patient from participating in the primary trial

Introduction

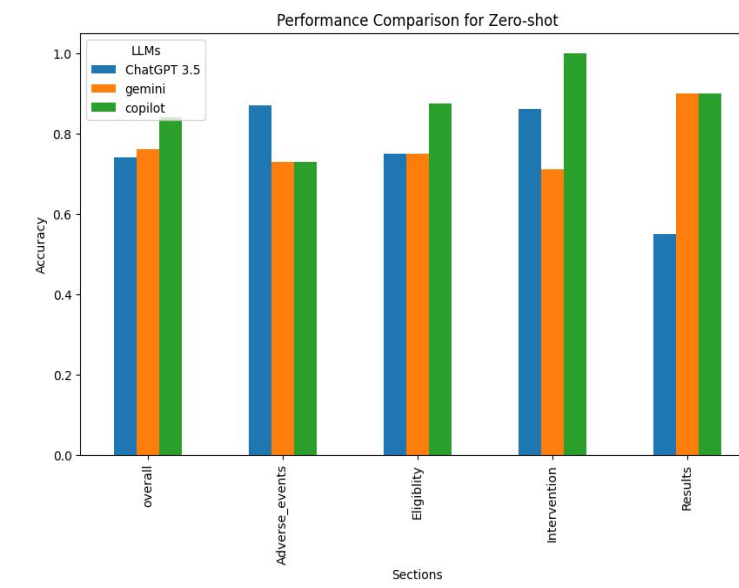
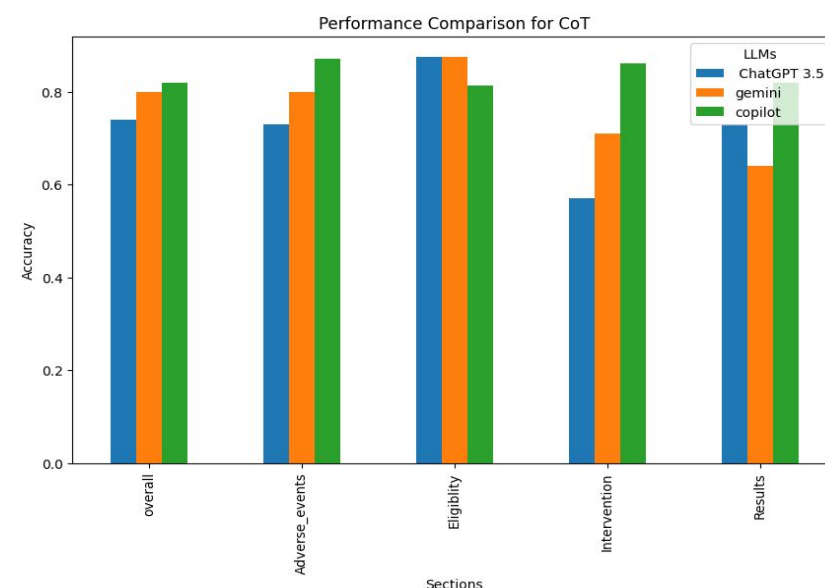
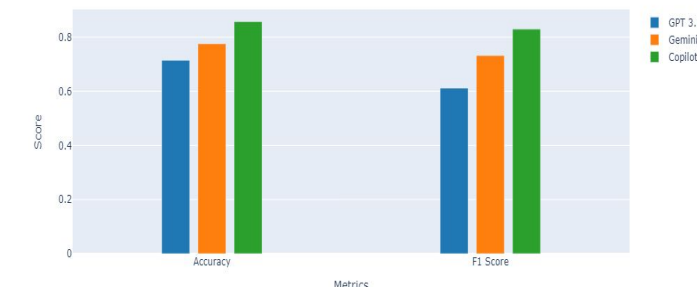


Experiments

Evaluation Metrics Comparison for CoT



Evaluation Metrics Comparison for Zero Shot



Evidence Retrieval

Model	F1 score	Precision	Recall
BERT	78.23	77.5	80.8
ClinicalBERT	81.1	81.0	81.2
BioBERT	80.4	78.0	81.9
multi-qa-mpnet-bas e *	52.9	53.2	52.6

*sentence BERT, cosine similarity

Conclusion

- Among the various LLMs evaluated, the GPT-4 based Copilot consistently outperformed others, yielding the best results.
- We found that on average CoT performs better than Zero-shot implying the CoT prompting strategy promotes step-by-step reasoning in Large Language Models (LLMs), which helps them break down the task into smaller, more manageable steps. This explicit reasoning likely enhances LLMs' understanding of Clinical Trial Reports, facilitating accurate inference of hypotheses' alignment with provided evidence.
- we observed that the performance is varying across the sections. Optimizing the prompts specific to each section may give better results.